Project Report : Full analog board

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1 Synthesis chain

In this section we are going to see the PCB in charge of sound synthesis. We will see its component, its structure and its use. There is four parts in sound synthesis: signal synthesis, summation and filtering. We have to add a fifth part to control each of them.

1.1 Signal synthesis

To create a complex sound we begin by generating various simple oscillations. To compromise liberty and size of the mixers, we choose to generate those simple signal with three Voltage Controlled Oscillator.

The first two call VCO A and VCO B are VCO 3340. They each generate three outputs at the same time. Those are saw, triangle and pulse signals. VCO A outputs are not all use together. Instead, a switch control by the user choose one of them. The last VCO is a 13700. It also generate triangle and pulse signal but also a sub-bass one.

To make sure that all of them are synchronize, all VCO must share their frequencies. For practical reasons, VCO A use VCO B frequency as inputs and VCO 13700 use VCO A frequency.

1.2 Signal synthesis

Once we have our seven signals we have to mix them. To exercise some control over this mix, every inputs is modulated by a gain chosen by the user. We decided to use two four entry mixers for this function. Signals are modulate with a VCA then convert with a resistance from voltage to intensity. Those intensities signal are simply connected to sum them before reconverting to a voltage output.

For practical reason, the first mixer call A have for inputs VCO B triangle, VCO A switchable output, VCO B saw and VCO B pulse. The second mixer call B have for inputs VCO 13700 sub-bass, VCO 13700 pulse, VCO 13700 triangle and VCO A switchable output again. The last one is not combine with the seven other. Instead it is only used to generate a information signal. The frequency is encode in the intensity for controlling the synchronisation of VCO 13700.

1.3 Signal summation

Only combining oscillator won't produce all the sound we wish. We use a configurable linear second or fourth order filter. This Voltage Controlled Filter can be Low-pass, Band-pass, High-pass or Notch. It only depends on the cut frequency, the resonance frequency and the order all chosen by the user. How the Multi-mode Analog VCF work is more detail in its own chapter.

1.4 Signal filtering

At this point, we have a custom filtered linear combination of a range of twelve oscillator. To make sure this synthesis won't hurt the listener, it is modulated with a Voltage Controlled Amplifier. The controller generate a customising ADSR envelope for the VCA. The signal generated will sound like a real note.

We can now send this sound to an audio output like a Jack female adapter or another card of the synthesizer.

2 Design the assemblage

We have seen all the individual parts generation the signal. We now need to combine and control them.

2.1 Control block

To control every parts we need an micro-controller cheap with enough outputs for:

- VCO A switch
- VCO A, VCO B and VCO 13700 frequencies
- VCO A and VCO B PWN
- Reset and Cutoff controls
- The eight mixer coefficients
- VCF two switches
- VCF parameters
- VCA envelop

We choose in the first place a Nucleo cheap STM32L432KC combine with an outputs expander MCP23017. But since most of those signal must be send analog and not digital, we must convert the cheap outputs. We choose a sixteen Digital Analog Converter but change it to two eight DAC.

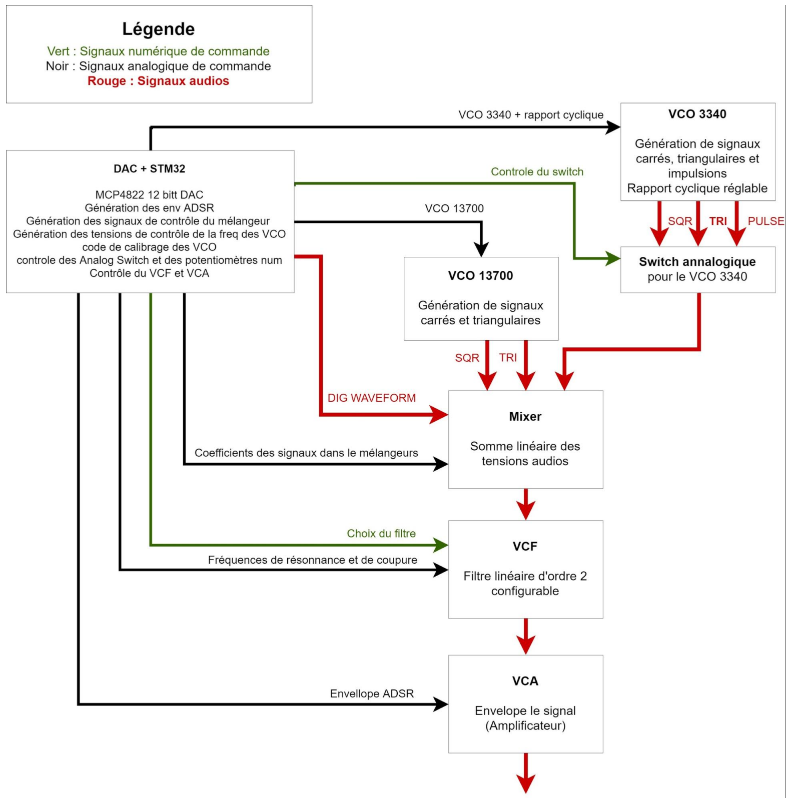

Figure 1: 'Functional diagram'

2.2 Schematic

To build the card, we first design the schematics. Each part schematics is connected to one another following the functional diagram.

2.3 Components choices

The next part diverge between the test board and the production board. We need to assign to each component a physical form. This mean choosing the right cheap for each linear amplifier and the series of capacitor and resistor to use.

For the testing board, we use capacitor and resistor passing through. This aloud to change them easily if we want to. For the production board we change to Surface Mount Device which are way smaller and directly weld on the board by the printer.

All capacitor does not have the same function. The majority of them can be made of ceramic for decoupling or bypassing but big decoupling capacitor (¿ 1 uF) need to be electrolytic and well oriented. Those related to timing must

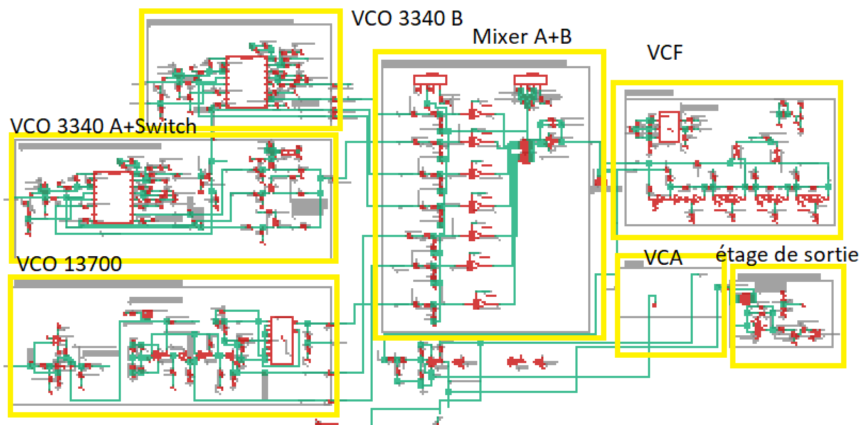

Figure 2: 'Schematics'

have great precision and will be tantalum one. They are located in the VCF, in the mixers output and in VCO timing control.

2.4 Board placement

The next step after the schematics and the choice of the components is the design of the board itself. In the first place we need to determine how each block will be place. Since the control block is link with all of them it must be centered. The mixers are placed above for practical details and readability. At their left VCOs.

The VCO B is the one with the most connection with the mixer A. Since it is also the smallest one is put on the top left corner. This let some space on the left for the second most connected VCO, the 13700 one, put under, on the left. Last but not least, the VCO A is place bottom to ease connection with the micro-controller, each mixer and the VCO 13700. To conclude and following the logical order, we place on the right the VCF, the VCA then the output stage.

3 Ordering

3.1 Security and test details

There is some minor aspect we have to check before trying to print this board. Since we will need to test every part of this board, we need to place a variety of voltage checkpoint and jumpers. They can be seen between every block. Two bottom are directly mounted on the board for testing the control.

Electromagnetic compatibility is something to be afraid of. Angle superior to 45 degrees is prohibited to limit resonance. The capacitor of each device must be as close as possible to it. This will insure a stable voltage. This is easier with SMD technology thanks to the capacitor size. The road are drawn as large

Figure 3: 'Block placement'

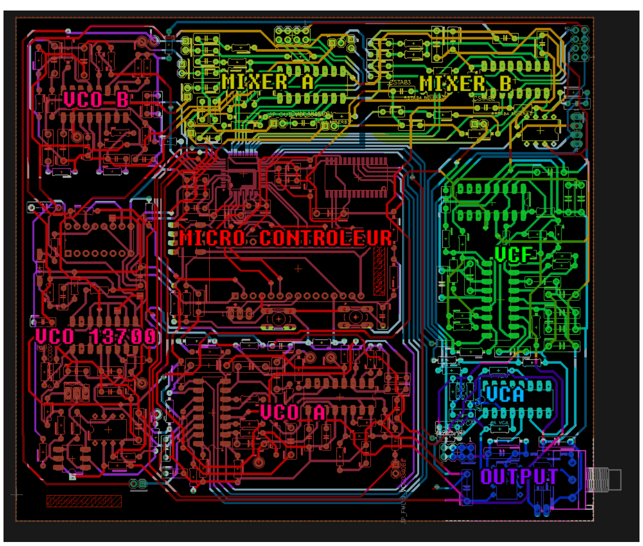

Figure 4: 'Board design'

as possible. A ground plane is extended on all possible surface top as well as bottom.

As the design of the Themis evolve, we need to build a board resilient to changes. At the outputs of the mixer and the VCA we have a connector to add other sources of sound. The micro-controller still have outputs usable and not connected. The last stage of the mixer B can still be use for other purpose thanks to a jumper.

3.2 JLCPCB order

We wanted to print this card at the ENSEA laboratory but its complexity and the recent pandemic force us to order it online. The Chinese company JLCPCB is know to be one of the largest and cheapest printer. To order them a board we need to follow three steps.

https://jlcpcb.com/parts list all the part we can ask them to directly weld on the board before send it. We need to make sure almost all of our components match theirs. By going to File / Export / Partlist in Eagle we can download our components sheet. We then need to modify our model index with JLCPCB assembly index found on their sheet.

https://jlcpcb.com/capabilities/Capabilities list what can be printed. We can also download a rule file readable by Eagle name jlcpcb2layer.dru to see if our board respect the capabilities. This is mostly an issue of size and minimal distances. Since the board was originally designed to be printable by the school lab, no modification were needed.

After creating a Gerber file version of our board, with File / CAM Processor then File / Open / Job on Eagle, we can finally order. No modification need to be done on the order sheet, just make sure the part list and the Gerber are update and loaded.